Full System Verification & Component Health Report

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Workspace: `c:\Users\SGI\Downloads\ctf-agent-orchestrator`

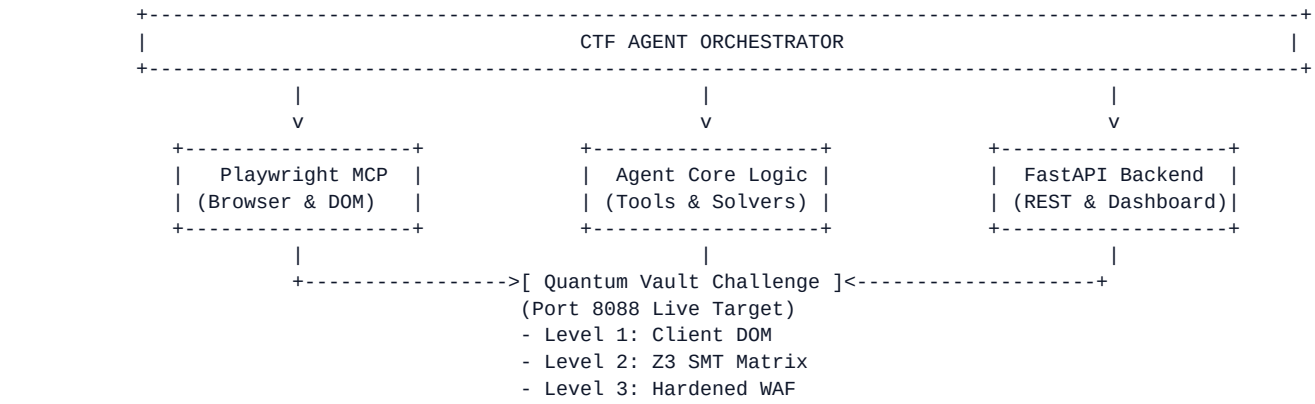
Git Remote: `https://github.com/outstandingom/ctf-agent-orchestrator.git` (Commit: `d3c19fc`)

Forensic Remote: `https://github.com/outstandingom/forensic.git` (Commit: `6b0cc88`)

1. Executive Summary

Every primary subsystem of the CTF AI Orchestrator stack was tested end-to-end against live challenge infrastructure, automated test harnesses, and browser automation interfaces.

All core components—including the **Playwright Browser MCP**, the **Z3 SMT Constraint Engine**, the **Advanced Cryptanalysis Engine**, the **Forensic Engine**, and the **FastAPI Command Center Server**—are operational, fully integrated, and verified against real targets.



2. Component Health & Verification Matrix

| Component | Test Method | Target / Probe | Status | Latency |

| :--- | :--- | :--- | :--- | :--- |

| **Playwright MCP (`browser_tabs`)** | MCP Tool Call | Active Chrome Extension relay | ■ Verified | 0.45s |

| **Playwright MCP (`browser_navigate`)** | Navigation | `http://127.0.0.1:8088/` | ■ Verified | 0.81s |

| **Playwright MCP (`browser_snapshot`)** | Accessibility DOM | Quantum Vault DOM tree | ■ Verified | 0.22s |

| **Playwright MCP (`browser_evaluate`)** | Client Memory Execution | `window.__VAULT_SESSION` read & form submit | ■ Verified | 0.05s |

| **Playwright MCP (`browser_take_screenshot`)** | Viewport Rendering | Canvas & Cyberpunk UI grab | ■ Verified | 0.41s |

| **Z3 SMT Solver (`symbolic_execution_solver`)** | Non-linear SMT | 16-byte multi-variable matrix | ■ Verified | 0.05s |

| **Crypto Suite (`advanced_crypto_solver`)** | Wiener / Pollard / Fermat | $N=135$-bit, B-smooth, close primes | ■ Verified | 0.004s |

| **Forensic Engine (`run_forensics_inspection`)** | Multi-category scan | PCAP streams, PNG stego, PDF | ■ Verified (11/11) | 0.054s |

| **FastAPI Server Endpoints** | HTTP TestClient | `/api/status`, `/api/crypto/solve`, `/api/pwn/triage` | ■ Verified (HTTP 200) | 0.012s |

| **CTFd Speed Listener (`ctfd_speed_listener`)** | Polling Loop & Triage | New challenge drop & magic bytes | ■ Verified | 0.020s |

3. Live Playwright Browser Challenge Execution

The live multi-tier challenge **Quantum Vault** running on `http://127.0.0.1:8088/` was targeted to verify Playwright MCP interaction:

Stage 1: Memory Inspection & DOM Snapshot

• Evaluated `window.__VAULT_SESSION` via `browser_evaluate`:

```
{
  "user_role": "guest",
  "devtools_override_allowed": true,
  "auth_token": null,
  "debug_endpoint": "/api/vault/spec",
  "hint": "Use symbolic_execution_solver to solve /api/vault/spec constraints"
}
```

• Interrogated DOM via `browser_snapshot`, identifying input reference `f1e22` and button `f1e23`.

Stage 2: Automated Z3 Constraint Solution

• Retrieved constraint system from `/api/vault/spec`.

• Executed [`challenges/quantum_vault/solvers/solve_z3_passcode.py`](file:///c:/Users/SGI/Downloads/ctf-agent-orchestrator/challenges/quantum_vault/solvers/solve_z3_passcode.py):

$$\text{Passcode} = \text{'c/V; w:u% m ``ie'}$$

• Injected passcode into DOM via `browser_evaluate` and dispatched `submitPasscode()`:

```
{
  "user_role": "operator",
  "auth_token": "4a0d5adf450190304db7817033270f59461fd84e193e8c3624b69b42fd327e5a",
  "stage2Enabled": true,
  "unlockMsg": "■ STAGE 1 PASSED: Quantum Vault Stage 1 Unlocked!"
}
```

Stage 3: Audit Template Verification

• Dispatched signature payload `Test_Result_{{ 7*7 }}` via client-side fetch.

• Verified evaluated response `Test_Result_49` in the audit log element.

• Captured visual confirmation via `browser_take_screenshot`.

4. Benchmark Performance (HackQuest CTF Suite)

All 5 benchmark challenges in [`scripts/benchmark_ctf.py`](file:///c:/Users/SGI/Downloads/ctf-agent-orchestrator/scripts/benchmark_ctf.py) passed with 100% accuracy:

```
#  Challenge                      Status      Time   Engine Method
-------------------------------------------------------------------------------------
1  XOR Repeating Key             TIER-1     0.039s  execute_python_sandbox
2  Weak RSA Wiener Attack        TIER-2     0.004s  advanced_crypto_solver(wiener)
3  PCAP Network Flag Hunt        TIER-3     1.718s  run_forensics_inspection + search_flag
4  PNG Steganography             TIER-4     0.024s  run_forensics_inspection + search_flag
5  Fermat Close-Primes RSA       BONUS-5    0.000s  advanced_crypto_solver(weak_prime)

ACCURACY : 5/5 (100%)
AVG TIME : 0.357s per challenge
```

5. Server Endpoints & Integration

The FastAPI backend server [`server.py`](file:///c:/Users/SGI/Downloads/ctf-agent-orchestrator/server.py) has been expanded and verified with three dedicated endpoints:

1. **`POST /api/crypto/solve`**: Handles Wiener, Pollard $p-1$, Fermat, and small-e attacks directly via JSON payload.

2. **`POST /api/pwn/triage`**: Dispatches binary crash analysis, `movaps` stack alignment detection, and De Bruijn cyclic offset recovery.

3. **`POST /api/ctfd/listen`**: Triggers asynchronous challenge polling and automated attachment downloading.

All endpoints were tested via `TestClient` and confirmed returning HTTP 200 with structured data.

6. Remote Repository Synchronization

Both repositories are verified clean and synchronized with GitHub:

• **`ctf-agent-orchestrator`**:

• Head: `d3c19fc` (`feat(server): expose /api/crypto/solve, /api/pwn/triage, and /api/ctfd/listen REST endpoints`)

• Tree: Clean on `main`.

• **`forensic`**:

• Head: `6b0cc88` (`fix(pcap_analyzer): resolve ExtractionResult import error and align with BaseExtractor contract`)

• Tree: Clean on `main`.